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Team 64: TI Solar Battery Pack Final Presentation

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Sponsor: Wyatt Keller
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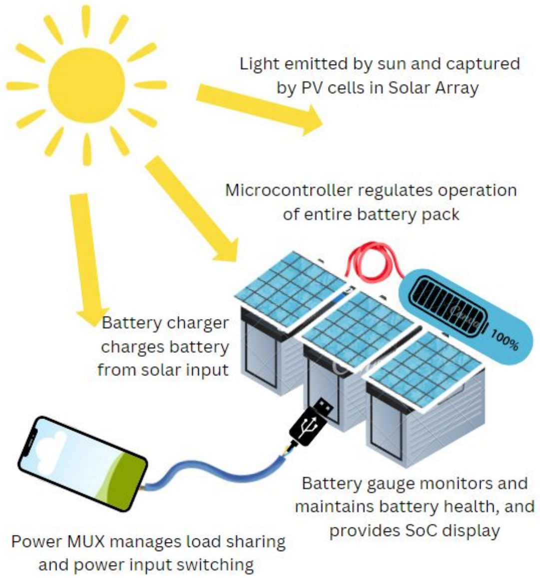
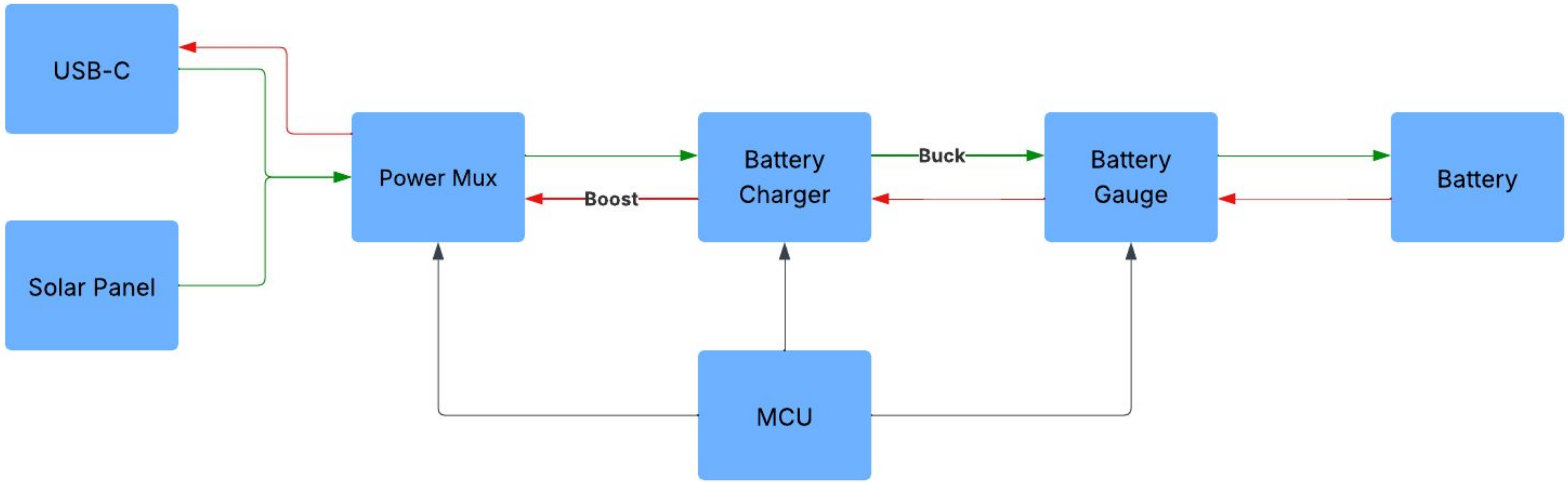
Project Summary

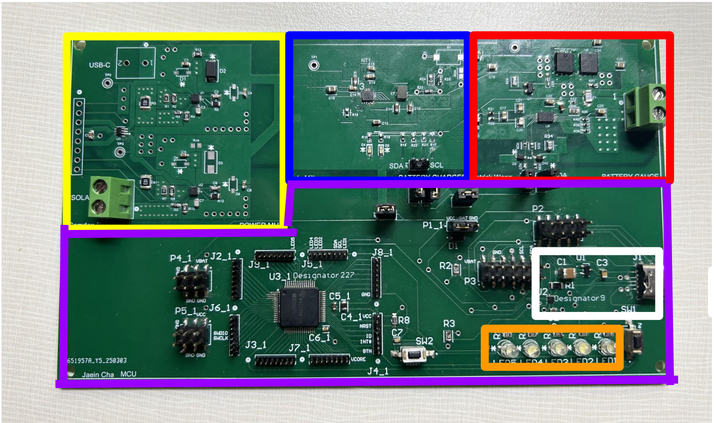
- Natural disasters, camping, or remote backpacking may result in a scenario with the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.



Subsystem Overview



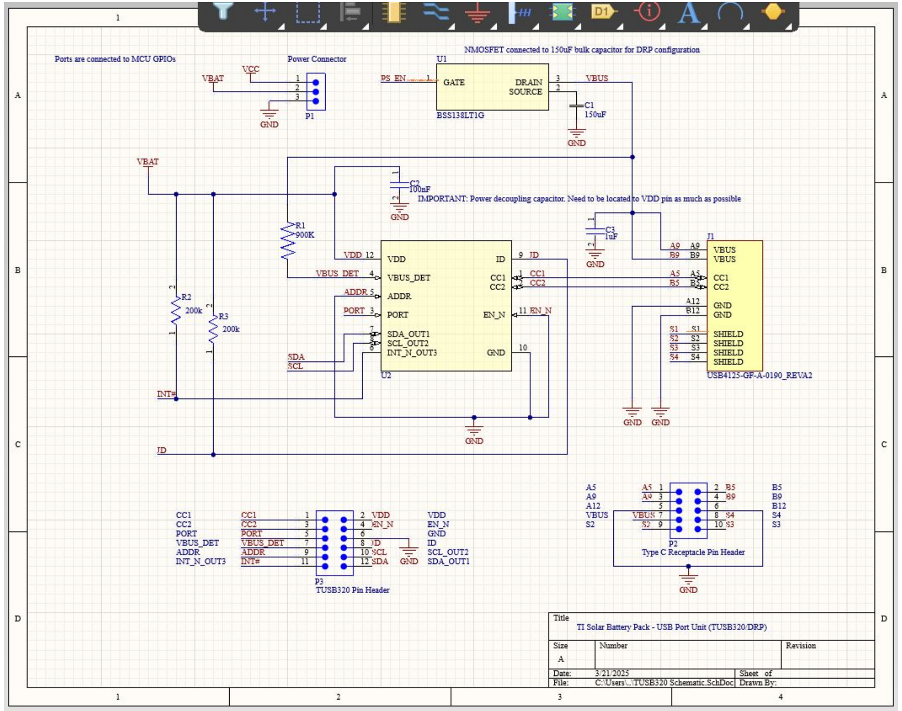
Integrated System Design



Yellow - Power Mux
Blue - Battery Charger
Red - Battery Gauge
Purple - MCU
Orange - SOC Display
White - USBC System

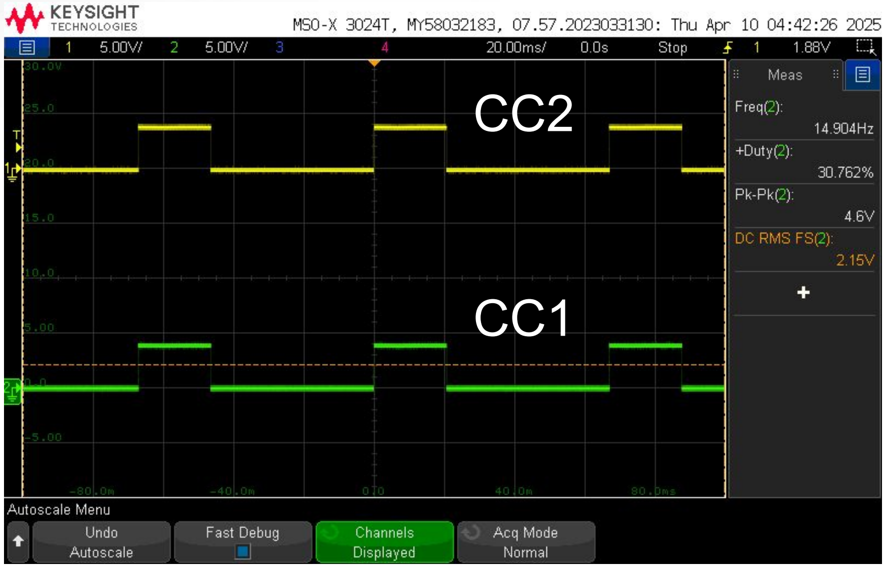
USB-C Input

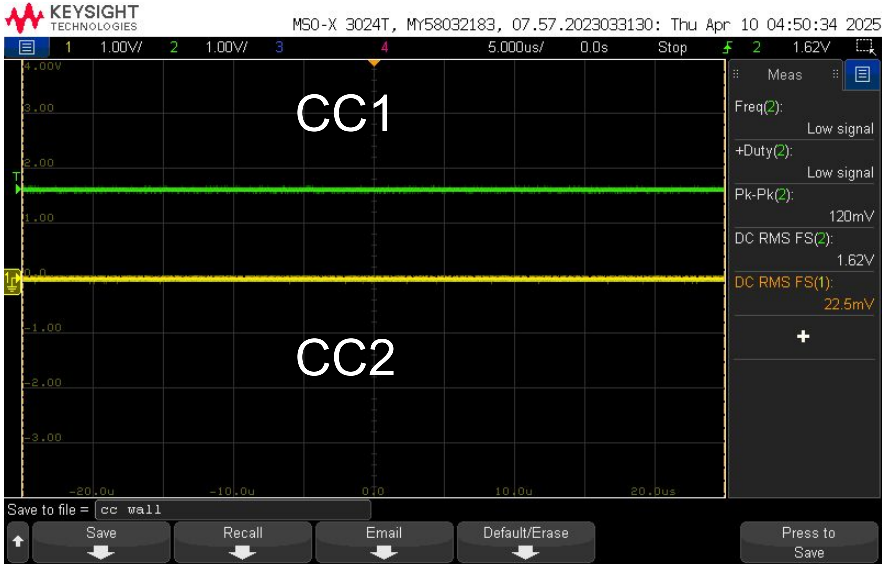
- USB-C logic controlled by TUSB320
- Configured as a DRP in I2C mode
- Validated to source and sink devices and control VBUS



USB-C Input

- CC lines when connected to a DRP (Left), and when connected to a source (Right)



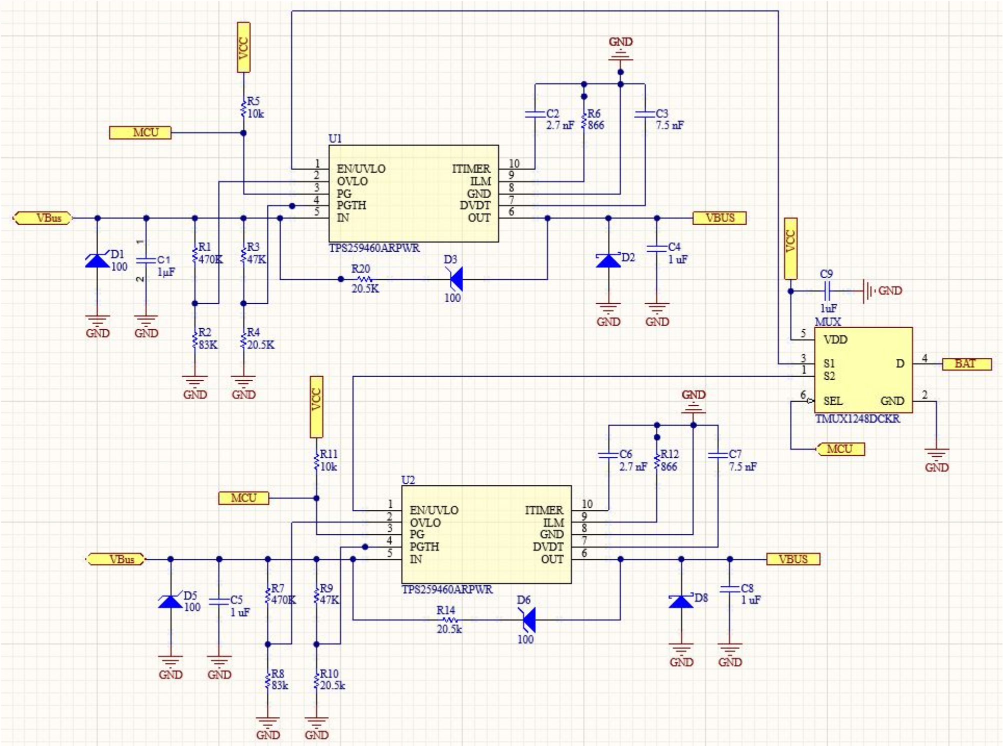


Solar Panel Input

- 2.5V to 7.5V output at 15W across three foldable lightweight panels
- On a standard sunny day charges at 7V 2A on average
- Cloudy day 7V at 0.5A

Power Mux - Camden Pomeroy

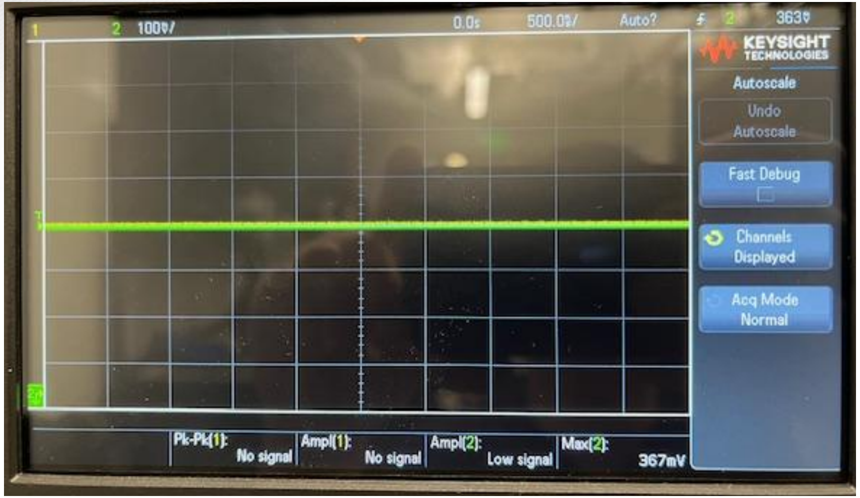
- Ensures protection of system when both USB-C and Solar inputs are connected
- Utilizes break before make switching
- UV/OV Shutoff Protection with Auto Retry
- MCU reading good input power detection and control over enabling power paths

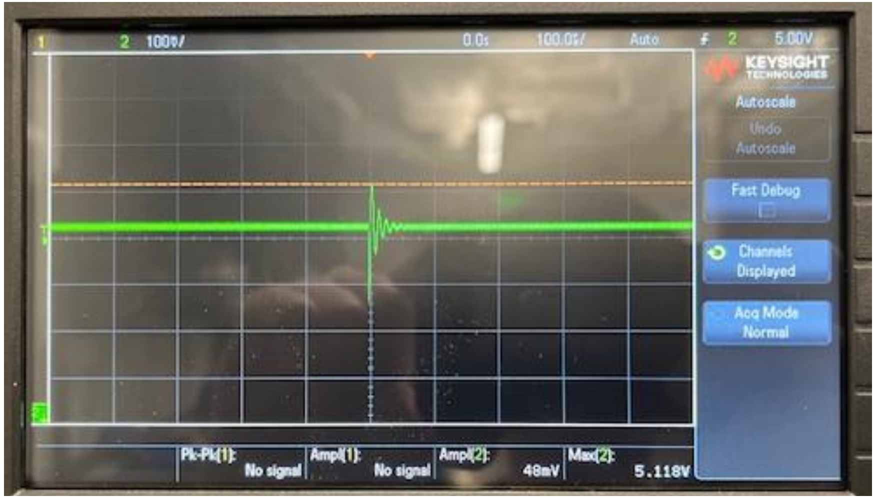


Power Mux - Challenges and Solutions

- High heat from high current paths
  - Created larger polygon pours and more stitching vias to boost efficiency
- Two loads would connect to VBUS at once sometimes
  - Increased MUX capacitor and redesigned drain and VDD pins
- UV/OV would trigger during expected operation
  - Gave a larger margin of error to account for MPPT of the solar panel on ideal and non-ideal days
- Powerflow only in one direction
  - Redesigned schematic to bias the IN side of the eFuse which allowed for Power to now flow from OUT to IN

Power Mux



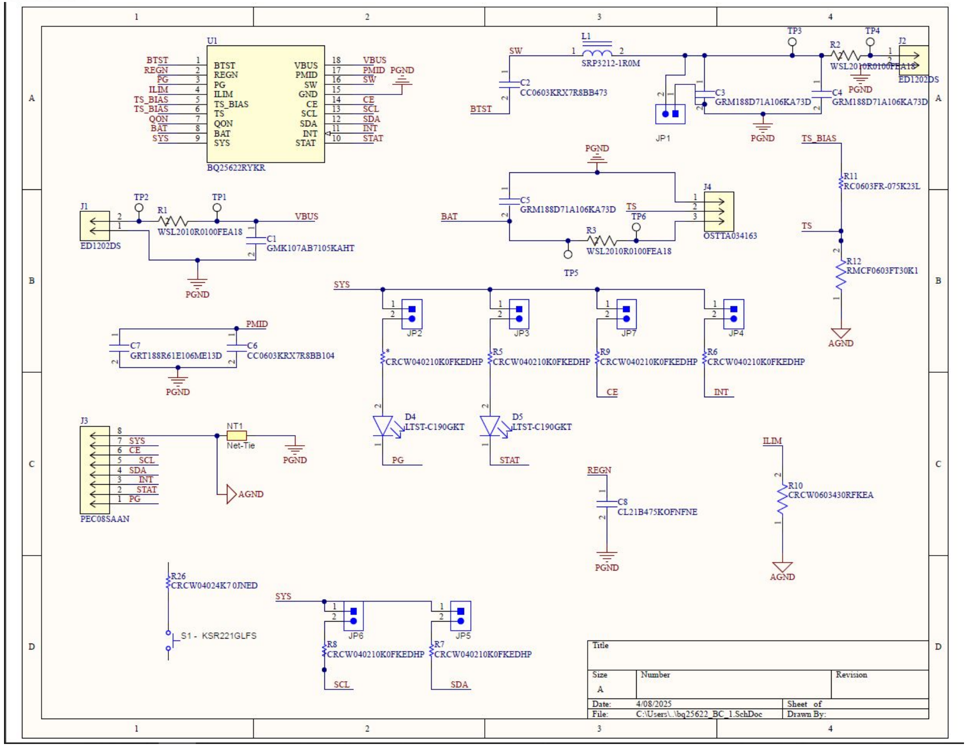


5V on USB-C and EN low on eFuse
~0V output

5V on USB-C and EN high on eFuse
~5V output

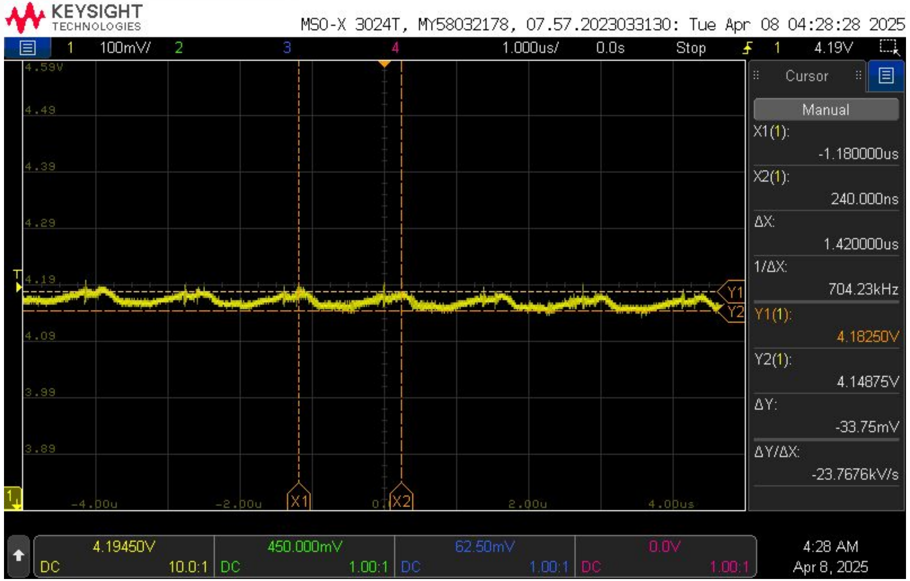
Battery Charger - Cole Miller

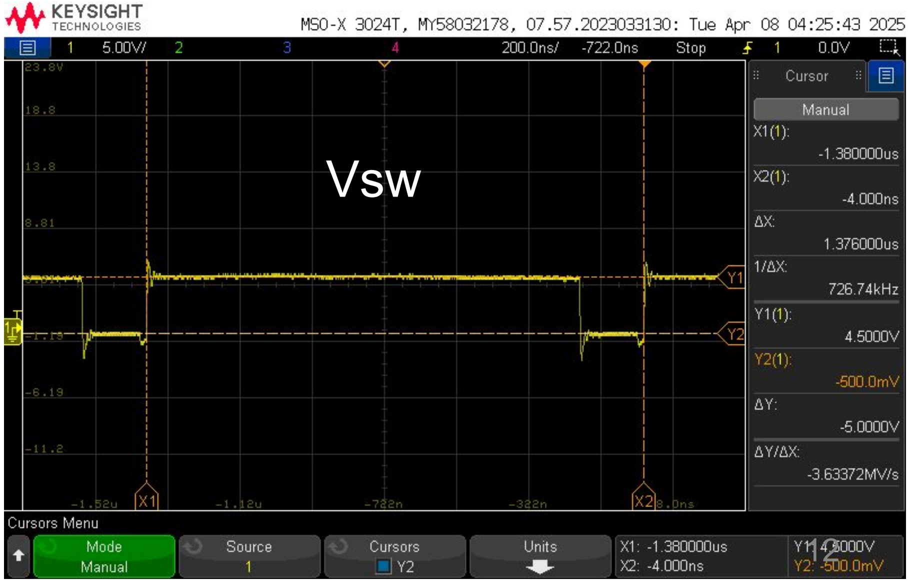
- Utilizes a BQ25622
- While charging battery acts as a buck converter
- While charging Devices acts as a boost converter
- Controls current and voltage



Battery Charger - Buck

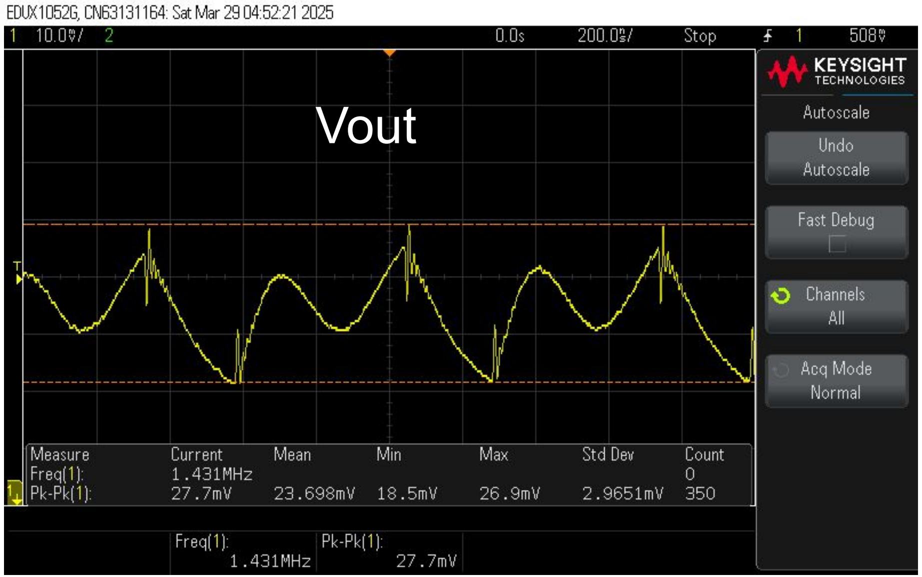
Depicted on the left is Vout at 4.2V with a 40 mV ripple. On the right is the switching voltage.

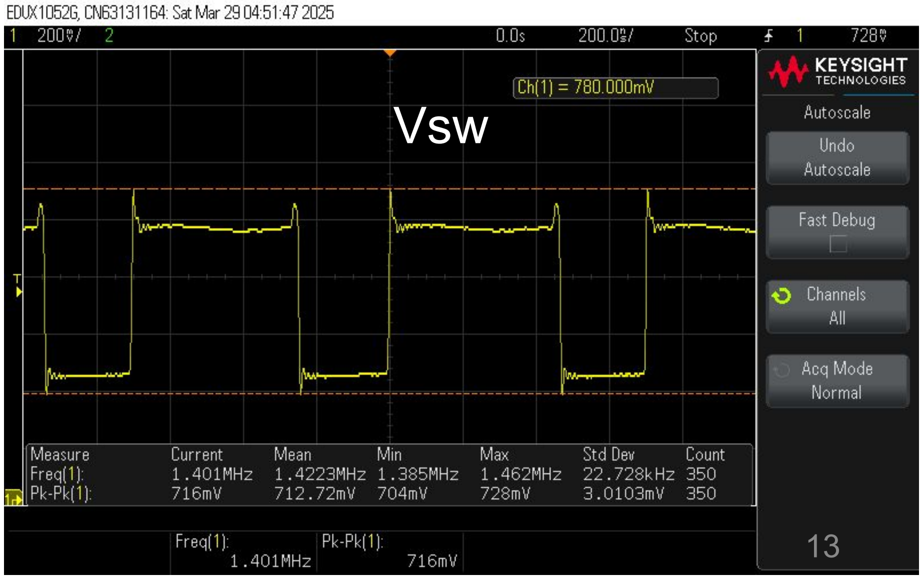




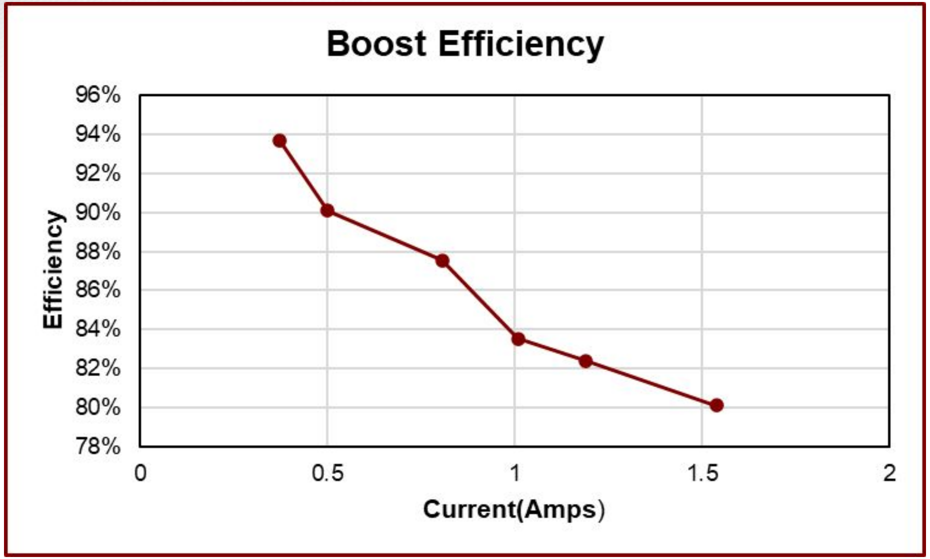
Battery Charger - Boost

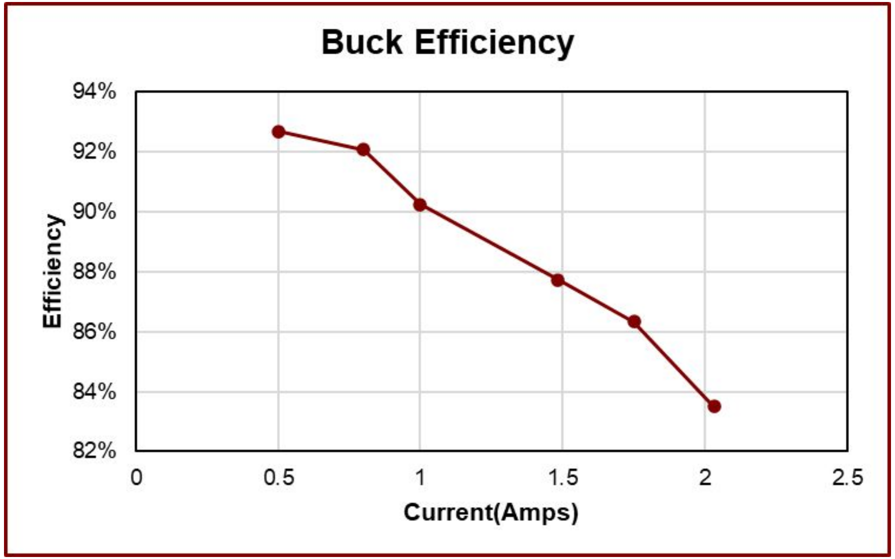
Depicted on the left is the AC portion of Vout at 5V with a 20 mV ripple. Depicted on the right is the switching voltage.





Battery Charger - Efficiency





Battery Charger - Challenges and Solutions

- High heat from high current paths
  - Created larger polygon pours and more stitching vias to boost efficiency
- Gate drivers wouldn't allow full voltage
  - Replaced ICs and changed soldering technique
- Lossy components due to ESR
  - Utilized ceramic capacitors and low ESR inductors
- Lack of knowledge on DC - DC converters
  - Utilized TI documents and forums to gain knowledge about package and operation

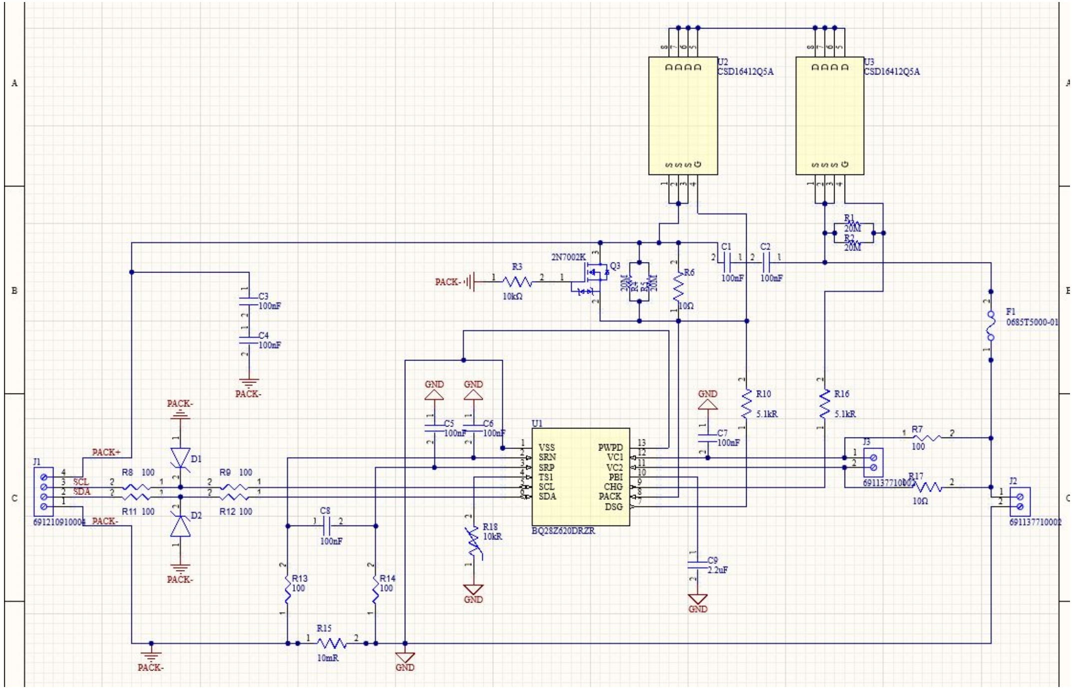
Battery Gauge - Patrick Wang

Overview

- Monitors voltage, current, temperature, and calculates state-of-charge (SOC) for Li-ion pack
- Configurable overvoltage (OV) and undervoltage (UV) thresholds to protect against faults
- Triggers fault flags (e.g., OV/UV, overcurrent) in registers

Challenges in 404

- Calibration of gauge to read proper charging parameters
- Extracting full charge cycle and discharge cycle data in one sitting



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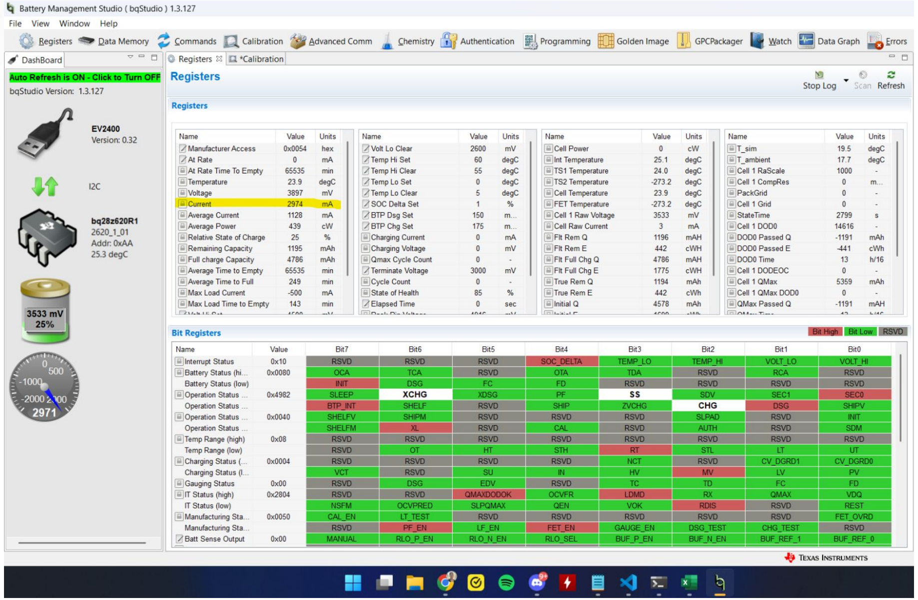
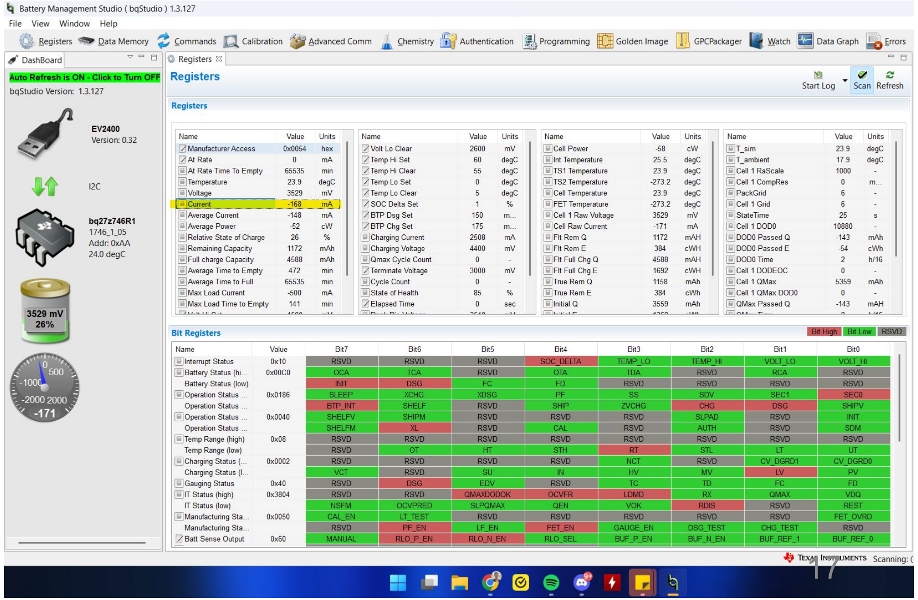
Battery Gauge Operation

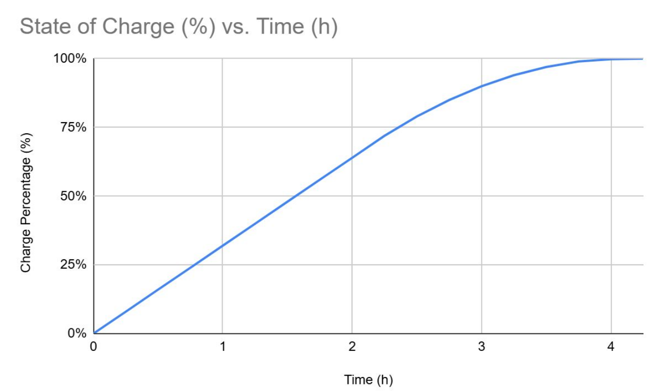
Input 4V 3A

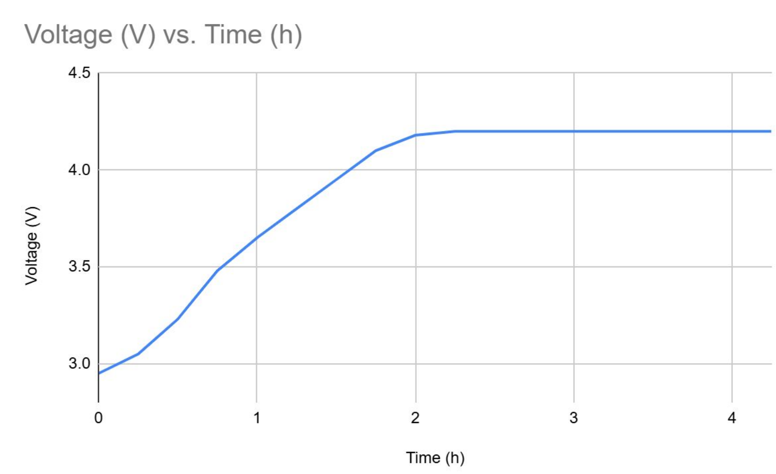


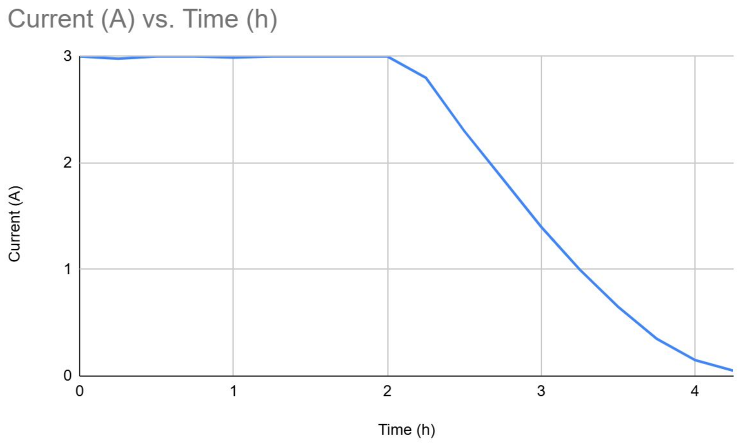
Input 2V 3A



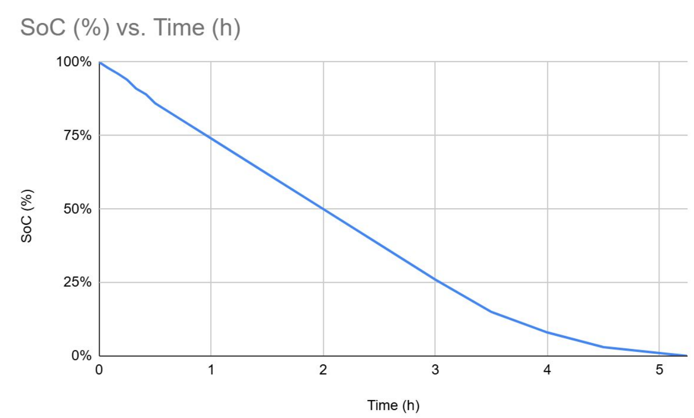
Battery Pack (charging)

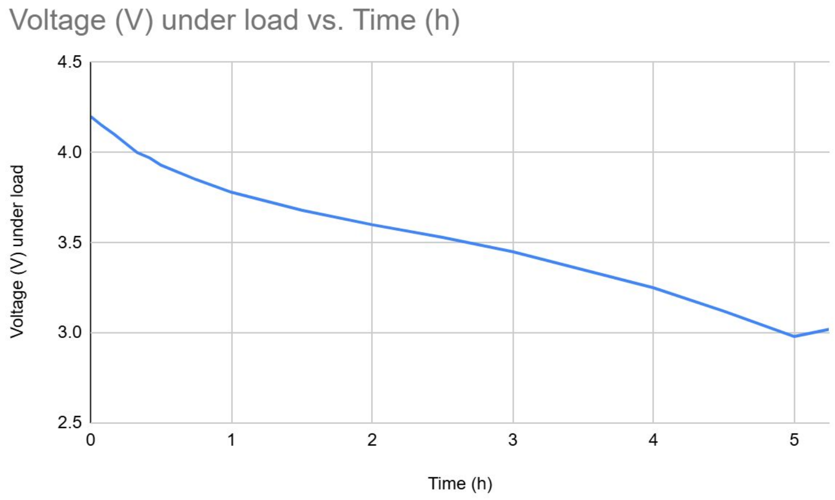


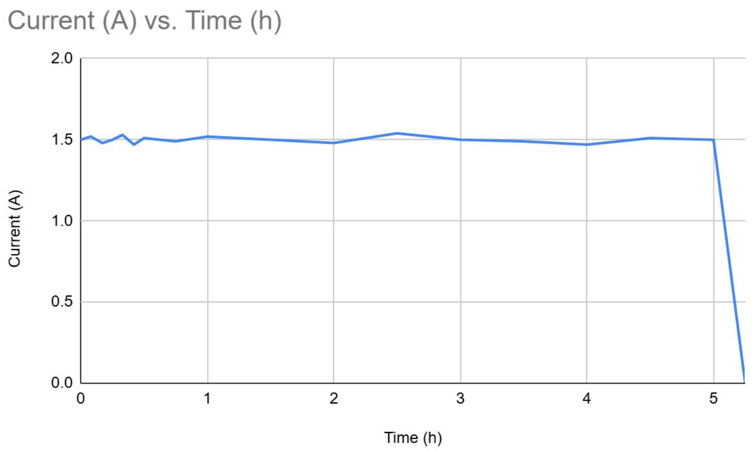




Battery Pack (discharging)



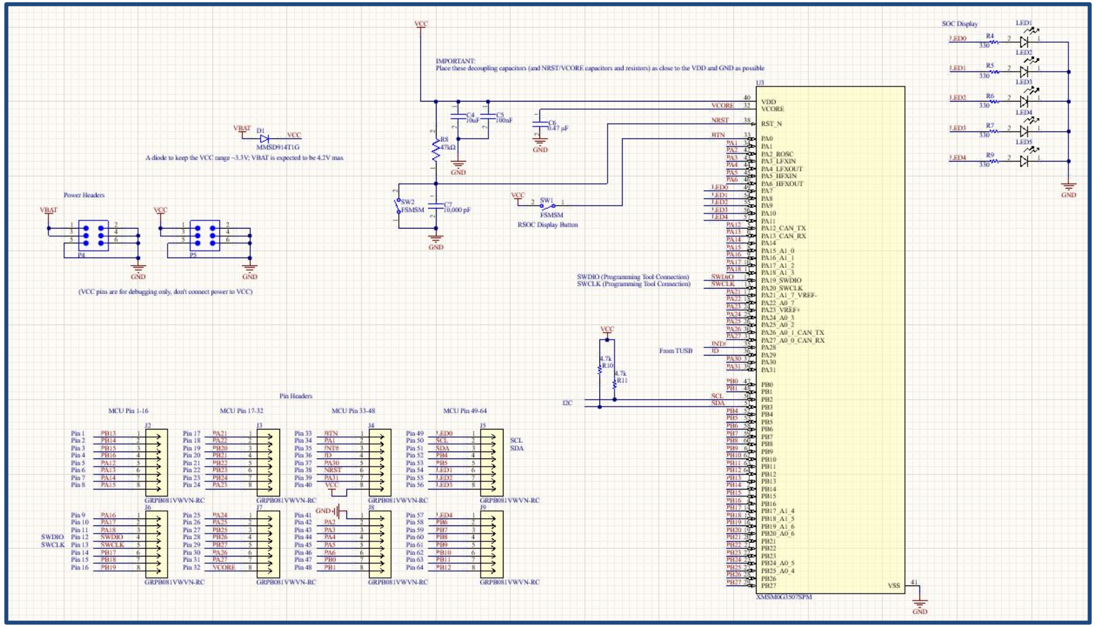




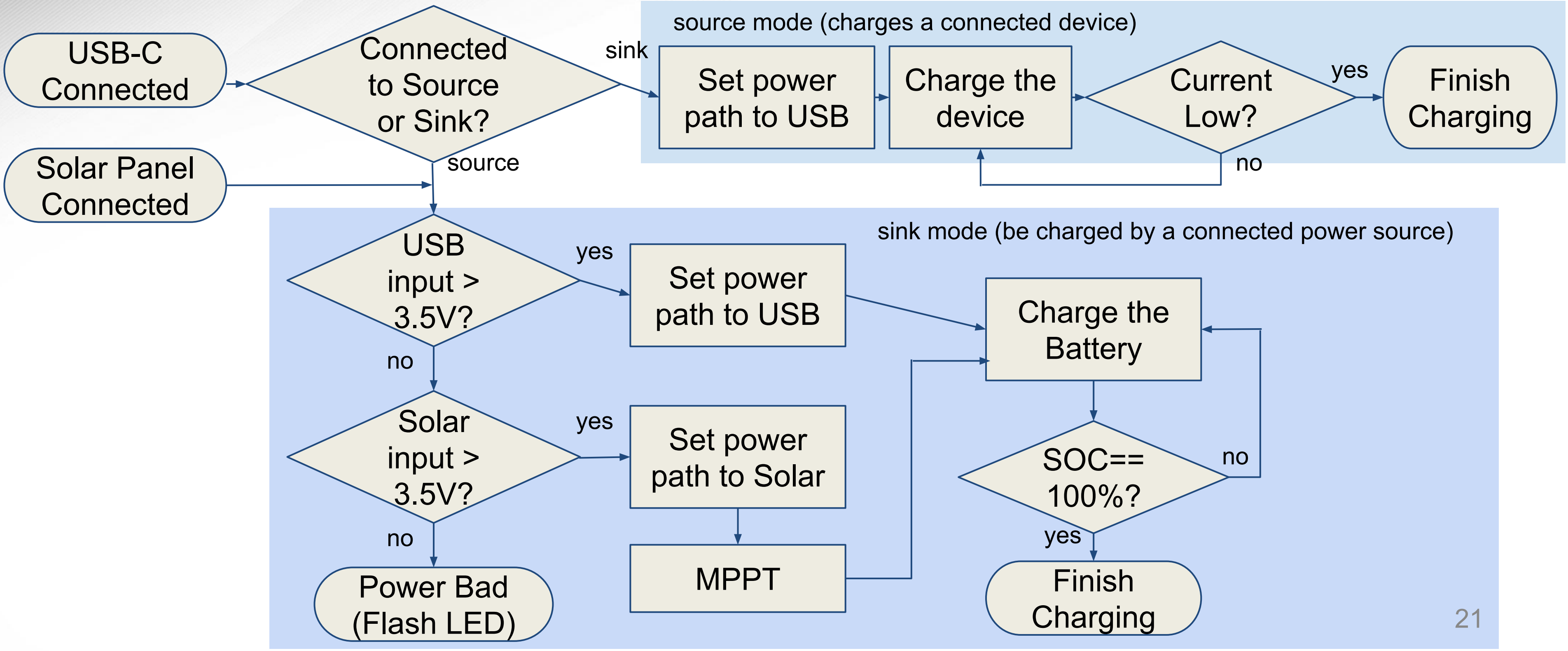
MCU - Jaein Cha

- Control unit - integrates subsystems and firmware into one organized system
- MSPM0G3507 (LQFP 64 pin)
- I2C Communication *Inter-Integrated Circuit*
  - Read/write register values for subsystem control
- MPPT *Maximum Power Point Tracking*
  - To maximize power from solar panel
- SOC display and SOC wake button

  SOC: State of Charge

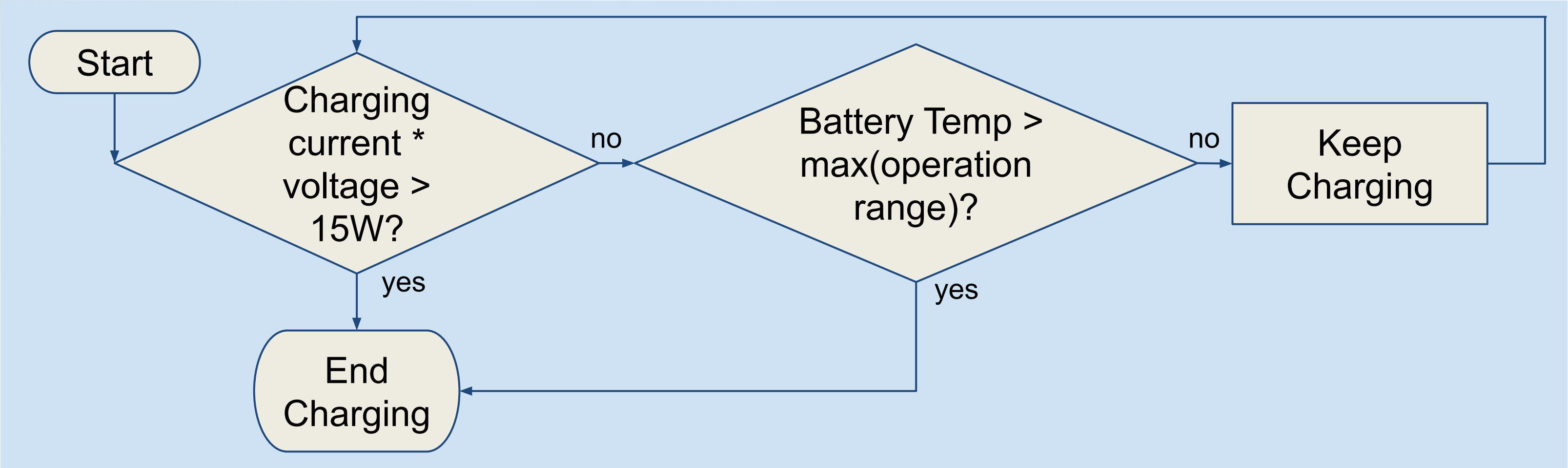


MCU Firmware Flow Chart

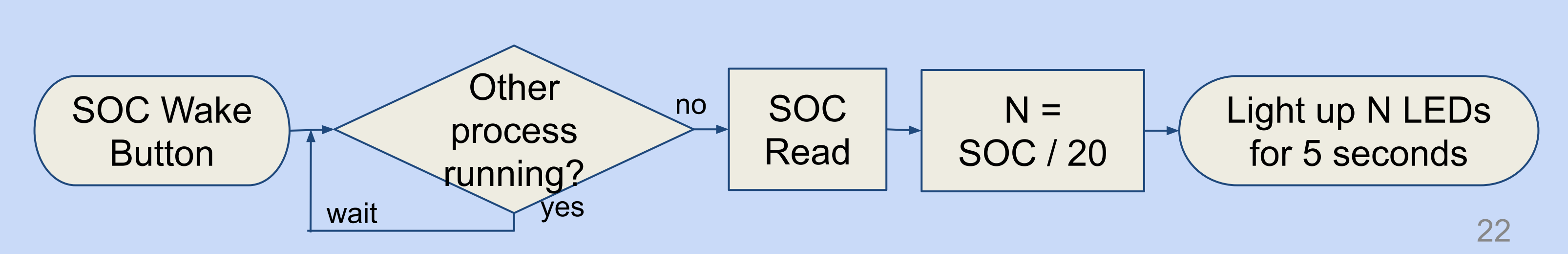


MCU Firmware Flow Chart

safety ensurement



SOC Display



MCU - Challenges and Solutions

- Inefficient process between firmware testing and on-board testing due to different package/pin configurations
  - Changed on-board MCU package to match with the one on the launchpad

- Firmware consists of different functionalities
  - Splitted firmware source code into various scripts, and assembled its functionality in the main.c file

- Complex to test due to its various functionalities
  - Tested each functionality separately prior to the integrated testing, checking the lists of interrupts and their priorities

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System Summary

Device	Charge Time
USB-C (5V 3A)	2 hrs 45 min
Solar Sunny (7V 2A)	3 hr
Solar Cloudy (7V 0.5A)	10 hrs 30 min

System Operating Mode	Current Consumption
Normal	1.2795 mA
Sleep	547.1 µA
Off	2.1 µA

Mode	Current	Discharge Time
USB 2.0	0.5 A	16 hr 40 min
USB 3.0 / 3.1	0.9 A	9 hr
USB BC 1.2	1.5 A	5 hr 15 min
USB-C Default	3 A	2 hr 30 min

- Batteries will last 16450 hours or about 685 days when device is idle in sleep mode

System Summary

Device	Battery Capacity (mAh)	Amount of Full Charges
iPhone 13/14/15	~ 3300	2.2
Google Pixel 5/6/7	~ 4335	1.65
Samsung Galaxy S23	~ 5124	1.85
AirPods Pro	~ 520	13.85
Nintendo Switch	~ 4310	1.67
Portable Fan / USB Light	~ 1200	6
GoPro Hero 10/11/12	~ 1720	4.2

Changes Since 403

- Integrated all individual subsystems together into one board
- Integrated new usb subsystem for input/output charging
- Changed MCU Package from VSSOP (28 Pin) to LQFP (64 Pin)
- Modified layout for battery gauge to reduce size

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System Validation Plan

Number	Validation Test	Input	Expected Output	Results	Status	Engineer	Reference Number
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	As expected		Camden Pomeroy	3.2.3.2.2
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected		Camden Pomeroy	3.2.3.2.2
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
7	PP1 UVLO Operational	if Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
8	PP2 UVLO Operational	if Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
9	PP1 OVLO Operational	if Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
10	PP2 OVLO Operational	if Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
11	SEL USBC	SEL is sent low from MCU	PP1 is active and PP2 is not	Not tested		Camden Pomeroy	6.1.4
12	SEL Solar	SEL is sent high from MCU	PP2 is active and PP1 is not	Not tested		Camden Pomeroy	6.1.4
13	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected		Camden Pomeroy	3.2.3.2.2
14	SOC Display	SOC from 0-100%	1,2,3,4, or 5 LEDs on as expected	Not tested		Jaein Cha	3.2.3.2.1
15	I2C functionality (1:1, EVM)	Run I2C test code	It reads the default register value	As expected		Jaein Cha	6.1
16	I2C functionality (1:1, on-board)	Run I2C test code	It reads the default register value	Not tested		Jaein Cha	6.1
17	I2C functionality (1:N, EVM)	Run I2C test code	It reads the default register values	As expected		Jaein Cha	6.1
18	I2C functionality (1:N, on-board	Run I2C test code	It reads the default register values	Not tested		Jaein Cha	6.1
19	I2C - initialization	3.3V Power supply to the po	Setup the communication correctl	As expected		Jaein Cha	6.1
20	I2C - DFP configuration	DFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
21	I2C - UFP configuration	UFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
22	I2C - DRP configuration	DRP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
23	Reset	Reset button pressed	MCU resets and stop working whi	Not tested		Jaein Cha	3.2.3.2.1
24	MPPT (simulation)	Various voltages	maintain the power as close as 15	As expected		Jaein Cha	6.2
25	MPPT (on-board)	Various voltages	maintain the power as close as 15	Not tested		Jaein Cha	6.2
26	SOP Display	SOP button	Lights up all the LEDs when SOP	Not tested		Jaein Cha	3.2.3.2.1
27	Standard Operation	5V 3A on VBUS	around 5V 3A on VBUS	4.2V 3.2A on VBAT		Cole Miller	2.1
28	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS		Cole Miller	2.2
29	ILIM Register Functionality	If input current >1A	output current >1A	Output 1A		Cole Miller	2.3
30	VLIM Functionality	If Vin > 4.2V	Vo > 4.2V	Output 4.2V		Cole Miller	2.4
31	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller	6.3.1
32	Battery Charger I2C	Write to registers. Write cha	Read from said registers	As expected		Cole Miller	6.1.2
33	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller	6.3.12
34	Battery Guage I2C	Write to registers. Read batt	Read from said registers	As expected		Patrick Wang	6.1.3
35	Battery Lifespan	Battery connected to load	The lifespan is at least 3 hours	As expected		Partick Wang	3.2.1.2
36	Power Consumption	Normal Operating Condition	Maximum power consumption is b	As expected		Patrick Wang	3.2.1.2
37	Battery SOH	Discharge batteries from 10	Batteries stay in within an appropi	As expected		Patrick Wang	5.1
38	High Power Path Charge	5V 3A from power MUX	Charges Battery Pack	As expected		Patrick, Cole, Camden	3.2.3.2.2
39	High Power Path Discharge	Load Connected to USBC	Battery Pack Discharges 15W	Not tested		Patrick , Cole, Camden	3.2.3.2.2
40	USBC Charging	USBC plug from wall outlet	Battery Pack Charges	Not tested		Patrick Wang	6.2.1

Questions?